Midterm Report: From Listings to Intelligence

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November 6, 2026

Abstract

Accolade, a property management platform, is developing a revenue management product to help property owners price their units correctly, so that they neither underprice their apartments nor overprice them and leave them sitting vacant. This report describes our progress on the first phase of that product, a market rent estimate. Because the real listing data was not yet available, we worked with a synthetic dataset provided by Accolade, which mirrors the structure of the expected data and, unlike real data, also records the true market rent behind every listing. We merged 11 of its 57 tables into a master table of 967,661 listings, separated the variables known when a unit is listed from those that reveal its price, and trained models on 2,400 buildings before testing them on 600 others. Gradient boosting predicts the asking rent with a median absolute percentage error of 10.6%, compared with 21.0% for a simple baseline. Since asking rent sits within 2.8% of the true market rent, most of the remaining error comes from information our variables do not yet capture, which motivates the comparable-property selection we plan next.

1 Introduction

1.1 Motivation and business problem

Accolade is a property management platform that is seeking to expand its product offerings by adding a revenue management product. The goal is to help property owners price their units correctly: an apartment priced too low leaves revenue on the table, while one priced too high sits vacant, and every vacant day is rent the owner does not recover.

Setting the right price depends on what comparable units are asking right now, and identifying that competitive set is an important challenge in itself. Geographic proximity alone may not be sufficient, since unit characteristics, amenities, building quality, concessions, fees, and other factors also affect how a unit is priced.

Throughout this report it is also important to distinguish three rents. The asking rent is the price posted on a listing. The effective rent is the asking rent less any concessions, such as a free month, spread over the lease:

$$\text{effective rent} \;=\; \text{asking rent} \;-\; \text{amortized monthly concession.}$$

The market rent is what a unit is actually worth given the market around it. Accolade's product aims to estimate the third from listings that only show the first.

Following Accolade's Draft Project Plan, the product is being built in two phases. The first, a market rent estimate, determines what comparable properties are currently asking for a given unit type. The second, a dynamic price recommendation, determines the asking price a property should post today based on its current price, demand relative to its occupancy target, comparable prices, price sensitivity, and recent leases. Phase 2 builds on Phase 1, since the recommendation uses the comparable set and the market rent estimate as its anchors.

1.2 Existing work

Rent estimation has long been approached through hedonic pricing, which treats the price of a property as the sum of the values of its characteristics [1]. To ground our approach in more recent work, each member of the team reviewed a paper covering a modeling technique, the effects of algorithm-based pricing, or the measurement of rental markets from listing data.

A study of residential property valuation in Germany compared an expert function for selecting comparable properties with hedonic models (OLS and GAM) and XGBoost, and found that XGBoost achieved the best predictive performance [?]. Work on automating real estate appraisals similarly treats comparable-property selection as the baseline that learned models must beat [?]. Other studies show the value of richer inputs: a Melbourne study combined text, image, and geospatial embeddings [?], a São Paulo study paired location-based features with natural language processing of listing descriptions [?], and a reproducible study of the major Spanish markets used machine learning to identify spatial market segments [?]. Finally, Zillow's home value index methodology shows how values can be measured consistently from listings over time [?], while a review of algorithmic pricing in the American rental market examined its effects with two-way fixed effects (TWFE) models [?].

These ideas map directly onto Accolade's data: automated comparable selection onto the platform's suggested comparables, image embeddings onto its listing photos and quality scores, and time-based validation onto its daily price series. They also shaped the models we consider, from hedonic regression to Random Forest, XGBoost, and stacked ensembles.

1.3 Problem statement

The technical goal of Phase 1 is to estimate the asking rent of a unit type, defined by its bedrooms, bathrooms, and a square-footage band, as a point estimate together with a range. Two requirements attach to every estimate. First, the range is required rather than optional, and it must be reported with the number of comparables or observations behind it. Second, every price must be explainable in terms of which comparables were used, which adjustments were applied, and which limits bound the result. Net effective rent is a derived second output rather than the first target, and it will be produced once concessions can be reliably extracted from listing text.

This report covers the Phase 1 modeling completed so far and the first component of Phase 2, the extraction of concession terms from advertised specials.

1.4 Overview

Figure 1 summarizes our pipeline. We merged the source tables of the synthetic dataset into a single master table with one row per listing, screened its variables for information that is not available when a unit is listed, scored the remaining variables, and trained and compared models on buildings held out from training. The generator's truth layer is used only to evaluate the predictions, never as an input. Section 2 describes the data, preprocessing, exploratory analysis, and methods; Section 3 presents our preliminary results; and Section 4 discusses our conclusions and next steps.

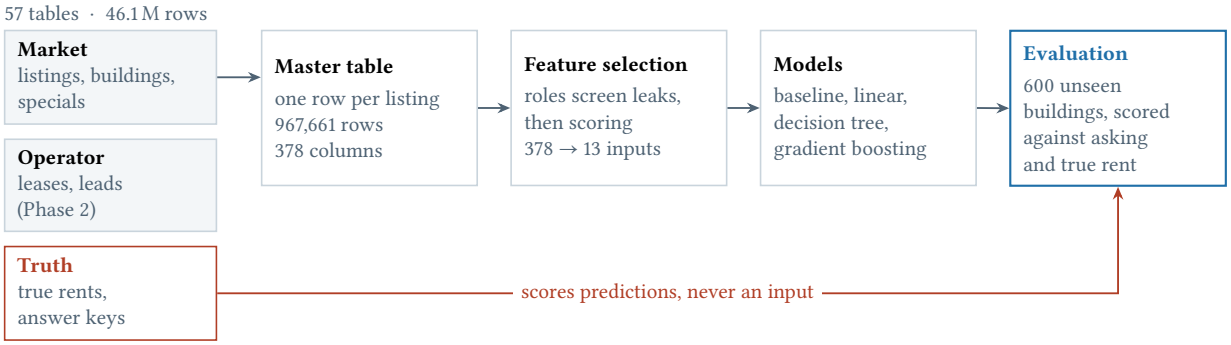


Figure 1. From source tables to evaluation. The generator's truth layer feeds only the evaluation: it scores the predictions and is never a model input.

2 Methods

2.1 Data

The original project proposal described a prepared seed dataset covering two to three U.S. metro markets, roughly 10,000 to 20,000 buildings and 500,000 listing observations, together with concession text, property photos, and analyst-curated comparable sets. During our first meetings, however, we learned that the real data was not yet available and that we would begin with synthetic data. Following Accolade's guidance, the synthetic data is used only for pipeline development and testing, not for model selection or tuning, and all of the analysis in this report is based on it. The real listing data has only just arrived, and we are now switching our work over to it.[1]

The synthetic data was built to conform to the structure of the real data: a market-relative dataset, which mirrors the Dwellsy listings feed, and a demand pricing dataset, which mirrors a property's own management-system records. It simulates 3,000 apartment buildings in three markets, Chicago (1,400), Dallas–Fort Worth (1,000), and Washington, DC (600), from 2020-01-01 to 2026-10-09, across 57 tables and 46.1 million rows. We organize these tables in three layers (Figure 1):

Market what a listings feed observes: buildings, floor plans, every listing with its dated price changes, and the free-text specials advertised with them.

Operator full property-management records for 150 buildings, including leases, guest cards, and a reference system's daily price recommendations, which we will use in Phase 2.

Truth the values the generator drew before anything was observed, such as each floor plan's true market rent by month, each listing's true lease, and each building's latent quality and pricing policy.

The truth layer is what makes the synthetic data especially useful. A real listings feed only records what landlords asked, whereas this data also records what each unit was worth, so we can evaluate whether a method recovers the market rent rather than whether it simply reproduces the listings.

By default, the generator describes an unremarkable, stabilized building with 12-month leases, fully priced and with nothing missing, and it then enables named edge cases on top. Attributes are masked: 12% of buildings are missing a year built and 29% a story count. 721 buildings are delivered partway through the history and lease up from empty, 43% of buildings reprice daily under revenue management while the rest adjust by hand, 1,688 vacancy spikes disrupt individual buildings, and 5 buildings in DC are rent-controlled.

[1] Accolade delivered a trial extract from Dwellsy, the listing provider, on October 22. It covers 975,917 listings of 263,688 rental units in the Chicago and Washington, DC metropolitan areas between September 2020 and October 2026. As advised, it is not used in this report.

Concessions appear in 91,156 advertised specials written in varied prose. Because each edge case is named, the generator can double as a test harness for the data-quality problems we expect in the real data.

One caution applies throughout. Because the generator builds rent from building attributes, a flexible model can score well partly by recovering the generator's own formula. Therefore, we treat our synthetic results as evidence that a method works when the data behaves as assumed, not as a forecast of its accuracy on real listings.

2.2 Preprocessing

With three markets and millions of rows, our first challenge was deciding how to examine the data at all. We began by merging the 11 source tables relevant to rent prediction into a master table with one row per listing, 967,661 listings by 378 columns. Building and floor-plan attributes join one-to-many, while a listing's price history and advertised specials are aggregated to the listing. Every join is checked for duplicated rows, and every column keeps the name of its source table (for example, `market_building__year_built`), so its origin stays clear after any selection. We then reviewed each variable's data type and content, and standardized the formats into strings, floats, integers, dates, and booleans for consistent use in later analysis.

A column's name, however, does not tell us whether it can be used. We therefore also tagged each column with a role (feature, label, latent, or leaky) and select inputs by role under a single rule: an input must be known on the day the unit is listed. Figure 2 shows why this rule has to be enforced by role rather than by correlation. Nine columns are the listing's own later prices and correlate with asking rent at $|r| \geq 0.987$, so they are easy to catch. Four more are the same prices per square foot, with $|r| \approx 0.57$, no higher than the best legitimate variable, so a correlation screen would keep them. Yet multiplied by square footage, they reproduce asking rent to within 1% for 100% of 925,373 listings ($r = 0.999991$). A model given these columns would score almost perfectly and learn nothing [2].

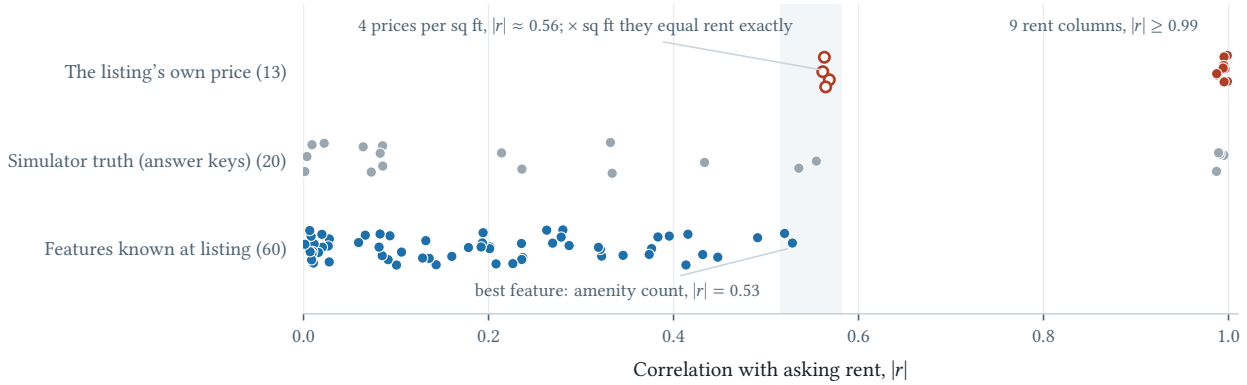


Figure 2. Correlation of every numerical column with asking rent, by role. Variables known when a unit is listed (blue) reach $|r| = 0.53$. The listing's own later prices (red) are leaks: nine rent columns are obvious, but four prices per square foot (open circles) fall in the shaded band beside the best variable, where correlation cannot tell them apart. Grey points are the generator's answer keys, never used as inputs.

2.3 Exploratory data analysis

Our exploratory analysis focused on how rent varies across markets, unit types, and buildings, and on which variables carry information about it. Median asking rent is $2,185 in DC, $1,820 in Chicago, and $1,426 in Dallas–Fort Worth (Figure 3). Between 2019 and 2026 it rose by 35% in Chicago and 25% in DC, while in Dallas–Fort Worth it peaked in 2024 and has since leveled off. Within each market, rent generally

rises with the number of bedrooms, but the spread within each bedroom count is wide (Figure 4), which suggests that location and building characteristics matter as much as the size of the unit.

Unit size shows why the building level matters. Across all listings, square footage has a Pearson correlation of 0.52 with asking rent, but within a single building the correlation rises to 0.84. In other words, size explains rent much better once the building is held fixed, because buildings also differ in many other ways that affect price. On the other hand, several variables that look promising are often missing: 4 fee and parking variables have Spearman correlations between 0.42 and 0.46 with rent, but are 70% to 89% missing.

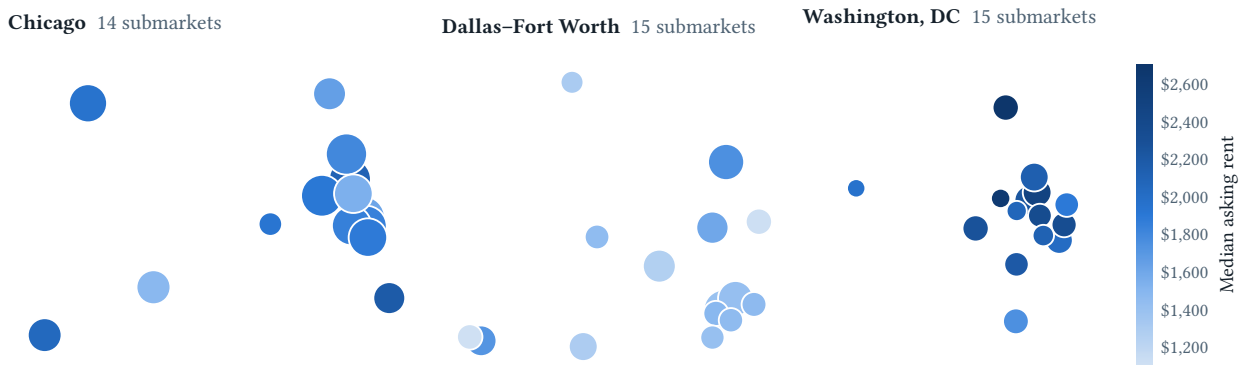


Figure 3. Median asking rent by submarket. Each circle is one submarket, sized by its number of listings and colored on one scale across the three markets.

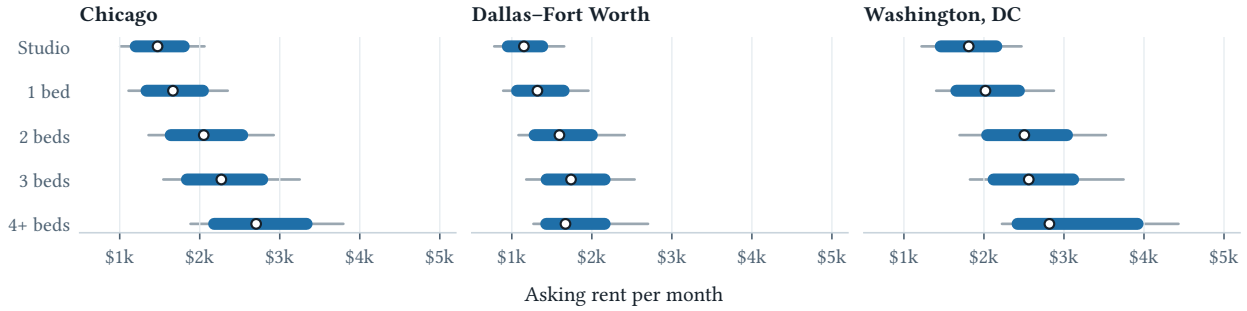


Figure 4. Asking rent by unit size. Dots mark the median, thick bars the middle half of listings, and thin lines the 10th to 90th percentile.

2.4 Feature selection

To narrow the 378 columns down to the model's inputs, we followed a staged funnel (Figure 8, in Appendix B). After removing keys, answer keys, and leaky columns, 317 columns remain, of which 299 are known when a unit is listed. We then analyzed the numerical and categorical variables separately.

For the numerical variables, we used two complementary approaches. The first scored every numerical variable with Gini, gradient boosting importance, and Lasso, and kept 16 of the 44 candidates. The second applied univariate, bivariate, and multivariate stages in sequence, narrowing 65 numerical predictors to 44 and then 18. The two selections agree on ten core variables, led by unit size (Spearman correlation of 0.51 with rent) together with location, the building's amenity count and review score, its leasing exposure when the unit was listed, and three measures of how often the building changes its prices. We also profiled the 215 boolean and 28 categorical predictors against the target.

For the categorical variables, we found that cardinality has to be judged per building rather than per row. A street address is only 0.31% distinct across listings but 99.97% distinct across buildings, so it is effectively a building identifier rather than a category. We therefore excluded identifiers and chose which categorical variables to keep based on their importance on buildings the model had not seen. Six of the 10 candidates are kept: property class, ZIP code, floor plan name, unit name, the predicted story bucket, and management company. On their own, they reach an R^2 of 0.55 on held-out buildings.

2.5 Models

Our models predict the first asking rent of a listing, the price posted when the unit comes on the market. As inputs we use 13 numerical variables known at that point: unit size, bedrooms, bathrooms, and floor; the building's amenities, review score, size, age, and application fee; its location; and the number of units sharing the floor plan. We compared three models: linear regression as an interpretable reference, a decision tree for its readable splits, and gradient boosting [3] to capture nonlinear relationships. Missing values are imputed with the median for linear regression and the decision tree, while gradient boosting, implemented as scikit-learn's histogram-based gradient boosting [4, 5], handles them natively. As a reference point, we also include a naive baseline that predicts the median rent for the listing's market and bedroom count. Appendix A lists the settings.

2.6 Concession extraction

As a first step toward Phase 2, we built a rule-based extractor for concession terms. Advertised specials are free text, such as "Enjoy the first month free on a 13-month lease," and Phase 2 needs their value in dollars to compute effective rent. We found that the concession type is already recorded in the market data and agrees with the truth layer for every one of the 91,156 specials, so only the value has to be extracted from the text. The extractor reads the amount, its basis (months free or dollars), and the lease term, and we score it against the truth layer.

3 Preliminary results

3.1 Evaluation

We evaluate every model on buildings it has never seen. Listings are split by building, 2,400 for training and 600 for testing, so no building contributes to both. With about 323 listings per building, a split by listing would let a model memorize buildings and overstate its accuracy. Our primary metric is the median absolute percentage error,

$$\text{MdAPE} = \underset{i}{\text{median}} \frac{|\hat{y}_i - y_i|}{y_i},$$

which reads directly as a typical miss, compares fairly across markets with different rent levels, and is not dominated by the few luxury units with the largest dollar errors. We also report the mean absolute error (MAE) in dollars and R^2. Additionally, we score each model twice: against the asking rent it was trained to predict, and against the generator's true market rent for the same floor plan and month, a comparison that only synthetic data allows.

3.2 Market rent

Table 1 compares the three models with the naive baseline. Gradient boosting performs best, cutting the baseline's median error by 50%, from 21.0% to 10.6%, with an MAE of $235 and an R^2 of 0.79 on the 198,589 listings of the held-out buildings. It also lowers MAE by 17% compared with linear regression. Because

the data are synthetic and a single test set was used to compare the models, these results are a working baseline rather than a model selection.

The last row of Table 1 is the most informative. A listing's own asking rent sits within 2.8% of the true market rent, which means asking rent is a reliable training target, and each model's error against the true rent (10.6% for gradient boosting) matches its error against the asking rent. The remaining error therefore does not come from noise in the listings. It comes from information about each building that our 13 variables do not capture, and comparable buildings nearby carry exactly that information, which is why comparable-property selection is our next step.

Table 1. Performance on 600 held-out buildings (198,589 listings). MdAPE is measured against each listing's asking rent and against the generator's true market rent for its floor plan and month. The last row is how far the listing itself sits from the truth.

	MAE	R^2	MdAPE (%)	
Model	($)		vs. asking	vs. true
Naive baseline (market × bedrooms median)	443	0.30	21.0	21.0
Linear regression	285	0.70	13.0	13.1
Decision tree	302	0.65	13.3	13.2
Gradient boosting	235	0.79	10.6	10.6
The listing's own asking rent	—	—	—	2.8

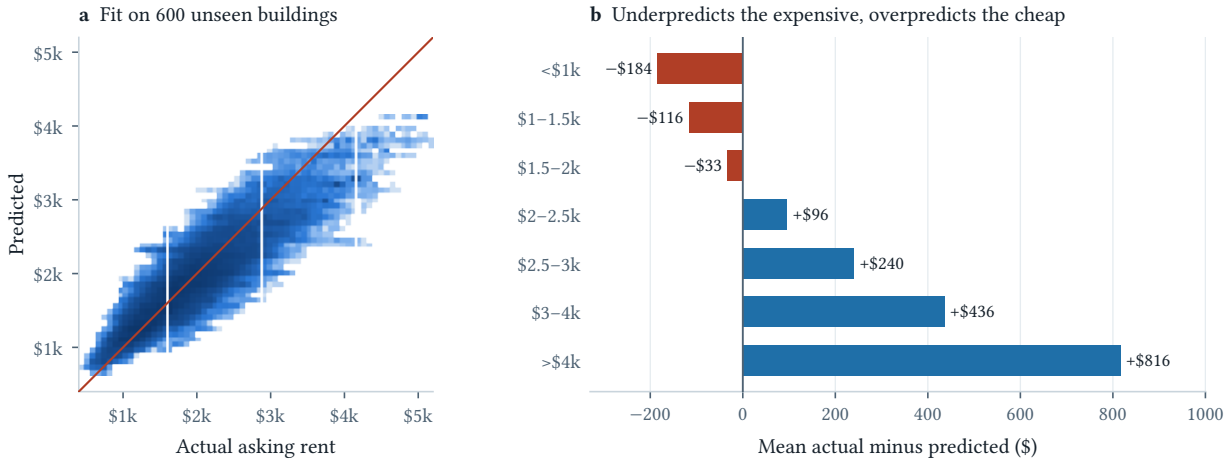


Figure 5. Predictions on held-out buildings. (a) Each square is a bin of listings, darker where more fall; the red line is perfect prediction. (b) Mean error by actual rent: the model pulls toward the middle, overpredicting the cheapest units by $184 and underpredicting the most expensive by $816.

3.3 Where the model errs

Errors grow with rent (Figure 5). The model pulls its predictions toward the middle: low rents are overpredicted and high rents underpredicted, by an average of $184 for units under $1,000 and $816 for units above $4,000. Figure 6 breaks the error down by segment. It is largest for class C buildings and for units with four or more bedrooms, and the bias differs by property class: class A buildings are underpredicted and class C buildings overpredicted. This pattern is consistent with the model not yet using property class, the strongest categorical variable we identified. Lease-up buildings, which we expected to be harder, are in fact predicted slightly better than stabilized ones (9.8% against 10.8%).

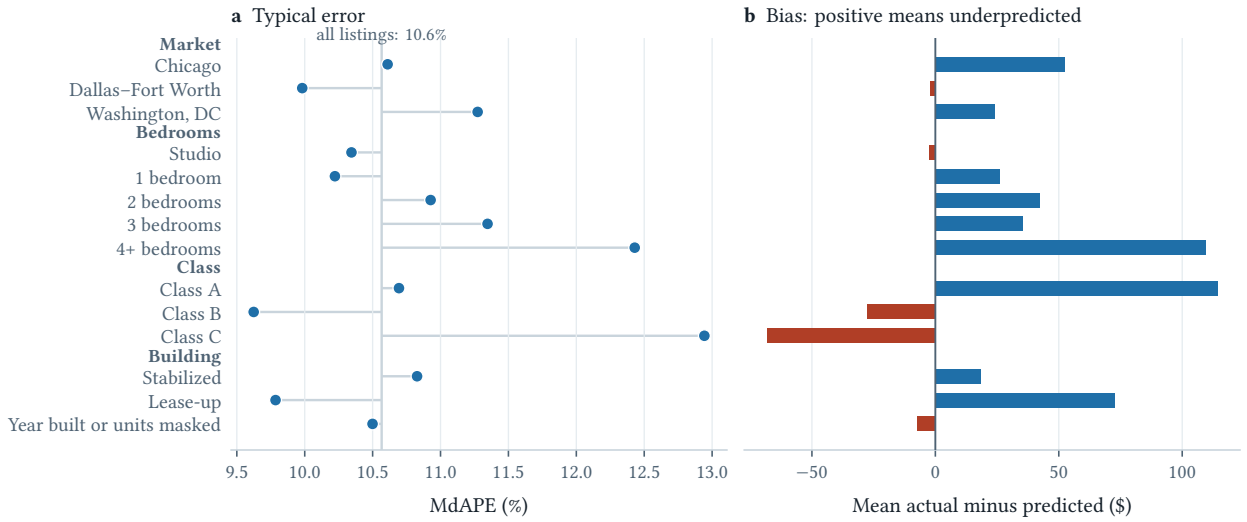


Figure 6. Error by segment on held-out buildings. (a) MdAPE; the vertical line is the overall 10.6%. (b) Mean actual minus predicted rent. Class A is underpredicted and class C overpredicted, consistent with the model lacking property class, the strongest categorical candidate in Figure 7.

3.4 What drives the predictions

Figure 7a shows how much each input contributes on unseen buildings, measured by the increase in MAE when that input is shuffled. Unit size matters most, adding $164 to the MAE when shuffled, followed by latitude, the building's amenity count, and longitude. Location ranks high in this test, consistent with our exploratory analysis, while bathrooms and the number of units sharing the floor plan add almost nothing once the other inputs are known. Figure 7b and c show why we select categorical variables on held-out importance rather than in-sample fit: state and metro area fit the training buildings well but add nothing on unseen ones, while property class is by far the most useful.

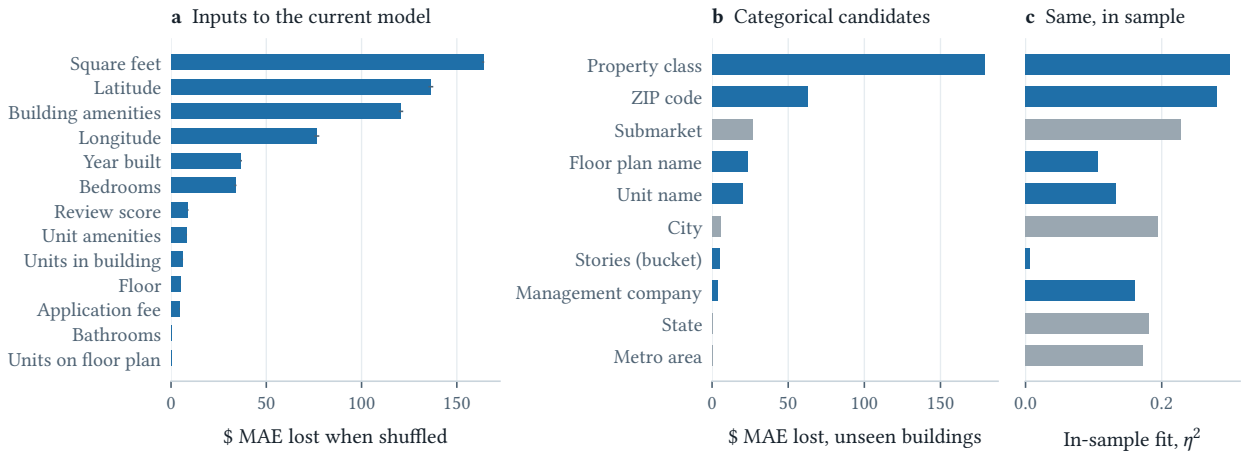


Figure 7. Which inputs matter on unseen buildings. (a) Dollars of mean absolute error added when each of the model's 13 inputs is shuffled. (b) The same measure for the categorical candidates, blue where kept. (c) Their in-sample fit: state and metro area fit the training buildings but add nothing on unseen ones.

3.5 Concession extraction

The extractor gets the amount right for 93% of the 77,429 specials that state their terms, and values the concession within 5% for 92% of them (Figure 9, in Appendix B). Accuracy depends on how the text was written. On specials written by a language model the rules are right 97% of the time, but on the generator's template text they are right only 62% of the time, because these specials often quote a dollar amount or a net effective rent where the truth layer records months free. Every dollar-denominated special is read correctly, while specials offering free months reach 87%.

3.6 Pipeline and reproducibility

All of our code and outputs are shared through the team's GitHub repository. Three commands rebuild the master table from the public download, and 59 join checks verify that the merged table is complete and consistent. This report is generated from the same code: every number in the text, every table, and every figure is recomputed from the data, so the report stays consistent with the models behind it.

4 Discussion

4.1 Conclusion

Working with synthetic data allowed us to build and test a complete pipeline for the market rent estimate before the real data arrived. We built a listing-level master table, protected it against leakage by selecting variables by role, and trained models on held-out buildings. Gradient boosting provides a working baseline that cuts the error of a simple benchmark by 50%. Because the synthetic data records the true market rent, we could also show that the remaining error is not noise in the listings but missing information about each building. Finally, we found that concession types are already recorded, so the work in concession extraction lies in recovering their value. However, these results cannot be considered representative of actual rental market conditions, and they need to be confirmed on real data. Accolade has now delivered the real Dwellsy listings, and we are moving our pipeline over to them.

4.2 Future Work

Our next steps build directly on these results:

- **Comparable-property selection.** Since the remaining error reflects missing information about each building, we will build the comparable-property baseline described in Accolade's plan, using similarity in location, quality, size, vintage, and amenities, and compare our models against it.
- **Combined variables.** We will merge the numerical and categorical variables under the same exclusion rules, starting with property class, and train with cross-validation grouped by property.
- **Additional models.** Beyond gradient boosting, we plan to evaluate Random Forest, XGBoost, and LightGBM, consider stacking based on their performance, and tune hyperparameters with Optuna [6].
- **Ranges and explanations.** Every estimate needs a range and its reasons, so we will add quantile or conformal prediction intervals [7], report their coverage, and explain predictions with SHAP values [8]. Accolade has also asked for an interpretable model, which we will weigh against predictive performance.
- **Evaluation.** We will report the median absolute percentage error by unit type and market, interval coverage, and performance on a later time period.
- **Phase 2.** We will continue with concession extraction and the demand-based price recommendation, for which we are considering survival models such as Random Survival Forests [9] and DeepSurv [10].

- **Real data.** With the real Dwellsy data now delivered, our immediate priority is to rerun the full pipeline on it, starting with Chicago.

Contributions

Maria Gonzalez Llama. Maria coordinated the team's weekly tasks and early communication with our mentor, prepared the preliminary note on project scope, and reviewed a German study on comparable-property valuation for the literature review. Maria developed the slides for our first capstone presentation, built the first model-training notebook with a leakage review and a grouped split, compared linear regression, a decision tree, and gradient boosting, and analyzed the errors by rent level, and has begun preparing the real Dwellsy data for modeling.

Viviana Hanampa Castillo. Viviana coordinated our first meeting with Accolade and the team's schedule, and reviewed a study of spatial market segments in the Spanish markets. Viviana wrote the Gini analysis and the report of string and numerical variables, scored all 108 numerical variables with Gini, gradient boosting, and Lasso, and built the correlation report, the decision table, and the exploratory notebook for numerical variables, and reorganized the repository by project phase, with notebook filters and join checks.

Shikha Mehta. Shikha reviewed competitor sites and multifamily rental websites to identify the variables we would need, and papers on multimodal models, Zillow's home value index methodology, and model evaluation for rent prediction. Shikha analyzed the numerical variables with Pearson and Spearman correlations, mutual information, within-building correlation, and permutation importance, wrote the plotting script for our exploratory figures, and found that size explains rent much better within a building than across all listings, and wrote several of our weekly reports.

Matthew Reynolds. Matthew researched listing data sources and established methods for rent prediction, set up the team's GitHub repository, and reviewed work on algorithmic pricing and TWFE models. Matthew was the first to work with the data, merging 11 tables into the 378-column master table. Matthew also built the categorical selection based on held-out importance, the Phase 2 concession extractor, and this report's figures and build pipeline, and handled communication with our sponsor.

Lady Infante Acosta. Lady reviewed a São Paulo study on rental pricing to identify geospatial features, and standardized the data types of every variable in a script that is part of our processing pipeline. Lady built the staged selection funnel with univariate, bivariate, and multivariate stages, narrowing 65 numerical predictors to 18, and profiled the 215 boolean and 28 categorical predictors against the target, and has begun the categorical analysis of the real Dwellsy data.

References

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A Model settings

All models use the 13 inputs listed in Section 2.5 and the same split: listings are grouped by building, and 20% of buildings are held out for testing (scikit-learn’s `GroupShuffleSplit`, seed 710).

Naive baseline the median asking rent of the training listings in the same market and bedroom count.

Linear regression ordinary least squares, after median imputation of missing values.

Decision tree maximum depth 10, at least 20 listings per leaf, after median imputation.

Gradient boosting histogram-based gradient boosting with 100 iterations, at most 31 leaves per tree, and a learning rate of 0.1; missing values are handled natively.

B Additional figures



Figure 8. From every column to the model’s inputs. Most columns known at listing time but not yet scored are the 219 amenity flags. The 22 kept so far are 16 numerical and 6 categorical.

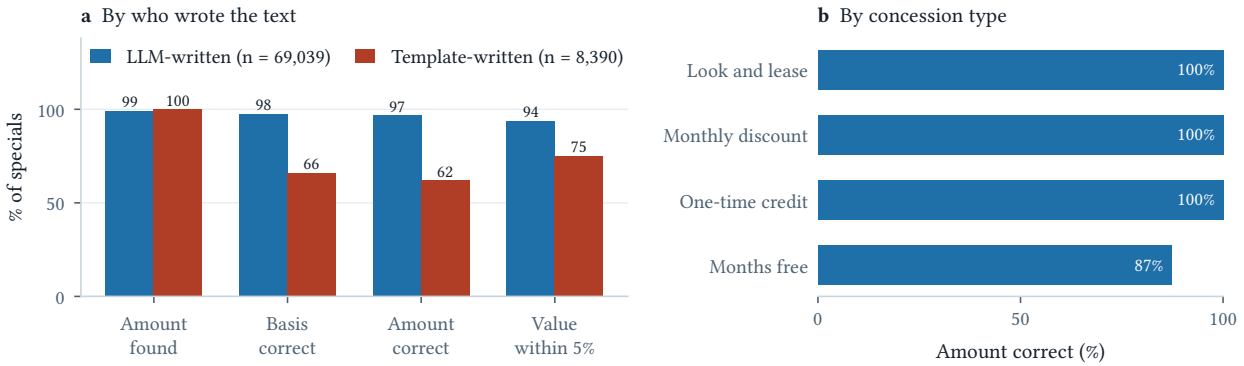


Figure 9. Rule-based concession extraction on 77,429 specials that state their terms. (a) Strong on text written by a language model, weaker on the generator's template text, which quotes dollars where the truth records months free. (b) Every dollar-denominated special is read correctly; months free reach 87%.